Supplementary material for "What the 50 Best Restaurants list can and cannot resolve: a sensitivity audit of its voting system"

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Section S1 lists the simulated rank interval of every restaurant of the 2026 list, and Section S2 reports the optimal-transport sensitivity indices of Section 6 of the paper. Code and data are at `https://github.com/Confareneoclassico/50best-voting-audit`.

S1 The 2026 list with simulated rank intervals

Table S1: Published 2026 position, 5th to 95th percentile of the simulated position under the published rules (pooled over the six worlds of Section 6), and the probability of a top-50 place.

Rank	Restaurant	City	90% interval	P(top 50)
1	Asador Etxebarri	Atxondo	1–1	1.00
2	Gaggan	Bangkok	2–3	1.00
3	Alchemist	Copenhagen	2–4	1.00
4	Quintonil	Mexico City	3–5	1.00
5	Wing	Hong Kong	4–9	1.00
6	The Chairman	Hong Kong	5–10	1.00
7	Kjolle	Lima	5–12	1.00
8	Kadeau	Copenhagen	5–12	1.00
9	Diverxo	Madrid	6–16	1.00
10	Plénitude	Paris	8–16	1.00
11	Atomix	New York	7–17	1.00
12	Sorn	Bangkok	8–22	1.00
13	Mérito	Lima	9–22	1.00
14	Sühring	Bangkok	9–24	1.00
15	Odette	Singapore	11–28	1.00
16	Tuju	São Paulo	10–30	1.00
17	Mingles	Seoul	11–30	0.99
18	Nusara	Bangkok	13–34	0.99
19	Jan	Munich	14–40	0.97
20	Piazza Duomo	Alba	14–38	0.98
21	Meet the Bund	Shanghai	13–45	0.97
22	Boragó	Santiago	14–52	0.93
23	Vyn	Skillinge	14–45	0.96
24	Al Gatto Verde	Modena	14–46	0.97
25	Lasai	Rio de Janeiro	14–50	0.95
26	Masque	Mumbai	18–56	0.91
27	Reale	Castel di Sangro	20–61	0.89
28	Trèsind Studio	Dubai	17–65	0.89
29	Enigma	Barcelona	15–68	0.89
30	Steirereck	Vienna	20–58	0.88
31	Celele	Cartagena	18–72	0.79
32	Potong	Bangkok	22–75	0.82

Rank	Restaurant	City	90% interval	P(top 50)
33	Elkano	Getaria	19–82	0.76
34	Le Coquillage	Saint-Méloir-des-Ondes	23–78	0.70
35	La Colombe	Cape Town	20–78	0.76
36	Ru Yuan	Hangzhou	21–73	0.72
37	Nuema	Quito	20–80	0.79
38	Atelier Moessmer Norbert Niederkofler	Brunico	23–98	0.71
39	Schloss Schauenstein	Fürstenau	24–86	0.60
40	Jordnær	Copenhagen	24–110	0.60
41	Onjium	Seoul	22–91	0.69
42	Bozar	Brussels	28–86	0.60
43	Le Calandre	Rubano	29–156	0.54
44	Don Julio	Buenos Aires	26–98	0.55
45	Mayta	Lima	22–86	0.55
46	El Chato	Bogotá	30–133	0.46
47	Le Louis XV - Alain Ducasse	Monte Carlo	29–137	0.47
48	Ikoyi	London	27–97	0.41
49	Naar	Kasauli	27–101	0.40
50	Saint Peter	Sydney	31–149	0.48
51	Uliassi	Senigallia	34–114	0.39
52	La Cime	Osaka	30–120	0.36
53	Ling Long	Shanghai	35–101	0.40
54	Nelita	São Paulo	34–127	0.29
55	Alcalde	Guadalajara	32–138	0.39
56	Sazenka	Tokyo	35–134	0.46
57	Koan	Copenhagen	36–169	0.23
58	Frantzén	Stockholm	30–162	0.27
59	Narisawa	Tokyo	36–177	0.25
60	Hiša Franko	Kobarid	32–173	0.28
61	Florilège	Tokyo	42–184	0.14
62	Fu He Hui	Shanghai	35–177	0.26
63	The Jane	Antwerp	38–186	0.20
64	Seroja	Singapore	40–189	0.17
65	Fyn	Cape Town	42–174	0.12
66	Willem Hiele	Oudenburg	39–167	0.19
67	Arpège	Paris	38–254	0.22
68	Evvai	São Paulo	42–257	0.19
69	Cocina Hermanos Torres	Barcelona	40–185	0.10
70	Kol	London	35–156	0.23
71	Salsify at the Roundhouse	Cape Town	42–266	0.12
72	Eatanic Garden	Seoul	41–196	0.14
73	Chef Tam's Seasons	Macau	42–238	0.16
74	Le Du	Bangkok	38–189	0.12
75	Belcanto	Lisbon	40–204	0.14
76	Nobelhart & Schmutzig	Berlin	37–168	0.14
77	Caprice	Hong Kong	44–269	0.09
78	Mosu	Seoul	42–231	0.09
79	Huniik	Mérida	47–254	0.14
80	Txispa	Atxondo	52–225	0.04
81	Septime	Paris	43–216	0.15
82	Coda	Berlin	51–277	0.04
83	The Ledbury	London	54–250	0.04
84	Le Doyenné	Saint-Vrain	48–260	0.07
85	Tim Raue	Berlin	44–216	0.11
86	Eliane	Brussels	43–323	0.08
87	Arca	Tulum	46–271	0.08
88	FZN	Dubai	54–281	0.03

Rank	Restaurant	City	90% interval	P(top 50)
89	Burnt Ends	Singapore	52–252	0.03
90	Baan Tepa	Bangkok	56–360	0.01
91	Den	Tokyo	52–247	0.04
92	Lamdre	Beijing	42–347	0.08
93	SingleThread	Healdsburg	48–319	0.08
94	Neolokal	Istanbul	52–324	0.05
95	Mil	Cusco	64–312	0.02
96	Rosetta	Mexico City	52–232	0.03
97	Khufu's	Giza	58–337	0.03
98	Labyrinth	Singapore	50–333	0.05
99	Fauna	Valle de Guadalupe	48–299	0.07
100	Aponiente	El Puerto de Santa María	61–306	0.03

S2 Optimal-transport sensitivity indices

The optimal-transport index of Borgonovo et al. (2025) compares the distribution of an output Y with its distribution at each level of a factor,

$$\iota_i = \frac{E[W_2^2(P_Y, P_{Y|x_i})]}{2\,\mathrm{Var}\,Y},$$

where W_2 is the 2-Wasserstein distance and the expectation runs over the levels of x_i, which are equally likely in the factorial design. For a one-dimensional output, W_2^2 is the integral over $(0,1)$ of the squared difference of the two quantile functions. We compute it exactly for the empirical distributions of the 6,912 simulated editions and of the 2,304 or 3,456 editions at each level. Writing m and σ for the mean and standard deviation of P_Y, m_i and σ_i for those of $P_{Y|x_i}$, and ρ_i for the correlation of the two quantile functions over $(0,1)$, the distance splits into three non-negative parts,

$$W_2^2(P_Y, P_{Y|x_i}) = (m - m_i)^2 + (\sigma - \sigma_i)^2 + 2\,\sigma\sigma_i(1 - \rho_i),$$

a shift in location, a change in spread and a change in shape. Averaged over the levels and divided by $2\,\mathrm{Var}\,Y$, the location part equals $S_i/2$, so $\iota_i \geq S_i/2$, with equality when a factor only shifts the output. An upper bound follows from coupling Y with the output Y' obtained by drawing x_i again while every other input, the panel included, stays fixed. Then $E[W_2^2] \leq E[(Y - Y')^2] = 2\,S_{Ti}\,\mathrm{Var}\,Y$, so $\iota_i \leq S_{Ti}$ for the total index of the model in which the panel draw is an input. The total indices of Figure 3, computed from cell means, leave out the interactions of a factor with the panel draw and can be smaller. A dummy factor that assigns the simulated editions at random to three levels gives the level of ι_i that sampling alone produces. Intervals are 95% jackknife intervals, as in Section 6.

Table S2 and Figure S1 report the indices and their parts. Where a factor matters for the rank correlation of the top 100 or the displacement of positions 51 to 100, the location part dominates and the index lies between the two bounds. The vote curve also changes the spread of these two features, which vary more across rules and panels when the top of the list collects fewer votes, and the allocation of voters changes the spread of the number of regions in the top 50.

For the four counts, the shape part is present even where a factor has almost no effect, and for the top ten it makes up most of every index. This is a property of counts that take few values. On the integers, moving probability p from one value to the next costs p in squared distance, while the mean moves by p, which enters the location part as p^2. A small change in the mean therefore appears mostly as a change in shape. The dummy factor shows the same pattern, with

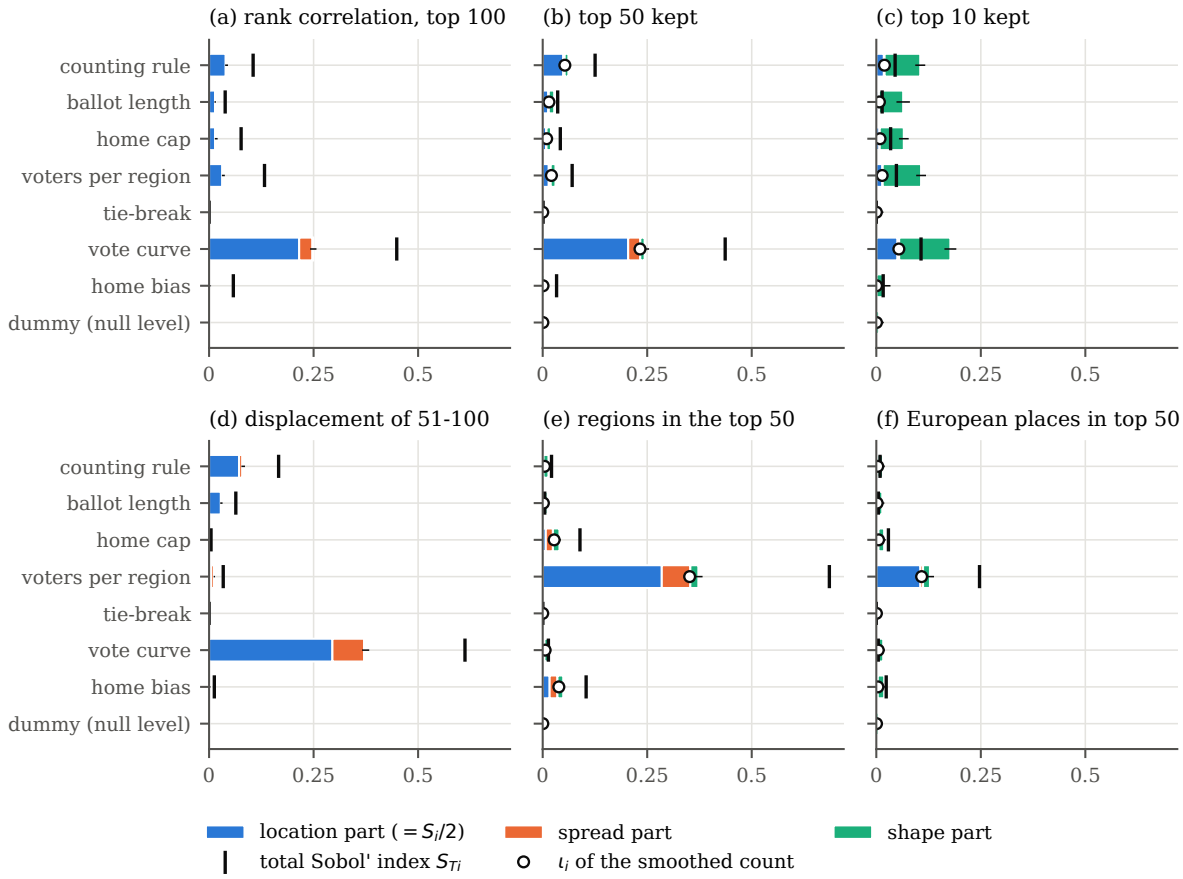


Figure S1: Optimal-transport indices ι_i of the six features of the simulated list, split into location, spread and shape parts, with 95% jackknife intervals. The location part equals $S_i/2$. Ticks: total Sobol' index S_{Ti} of Figure 3. Circles: ι_i of the counts smoothed by spreading each value uniformly over the unit interval around it. Dummy: runs assigned at random to three levels.

a null level of up to 0.007 on the counts against 0.001 or less on the two continuous features. In the last column of Table S2, each count is smoothed by spreading it uniformly over the unit interval around it before the index is computed. The shape parts then fall to 0.004 or less and the null level to 0.001 or less, and every index of a factor above the null level lies between $S_i/2$ and S_{Ti} of the smoothed count within its interval. The choice of index changes the size of the effects on the counts, then, but not which factors matter.

Table S2: Optimal-transport indices ι_i with 95% jackknife intervals, their location, spread and shape parts, the Sobol' indices of Figure 3, and, for the four counts, ι_i of the count smoothed by spreading it uniformly over the unit interval around it (last column). The bounds for the smoothed index are the Sobol' indices of the smoothed counts, which differ from the S_i and S_{Ti} shown by at most 0.011.

Factor	ι_i	95% interval	Location	Spread	Shape	S_i	S_{Ti}	ι_i smoothed
(a) Rank correlation of the 2026 top 100 with the published order								
Counting rule	0.042	0.038–0.045	0.040	0.001	0.001	0.080	0.105	–
Ballot length	0.015	0.013–0.016	0.014	0.000	0.000	0.029	0.039	–
Home cap	0.017	0.014–0.020	0.015	0.001	0.001	0.030	0.077	–
Voters per region	0.034	0.031–0.038	0.032	0.001	0.001	0.064	0.133	–
Tie-break	0.000	0.000–0.000	0.000	0.000	0.000	0.000	0.000	–
Vote curve	0.249	0.242–0.257	0.216	0.032	0.002	0.431	0.449	–
Home bias	0.004	0.003–0.005	0.002	0.002	0.001	0.003	0.058	–
Dummy	0.000	0.000–0.001	0.000	0.000	0.000	–	–	–
(b) Published top 50 kept in the top 50								
Counting rule	0.061	0.057–0.066	0.049	0.003	0.009	0.099	0.125	0.053
Ballot length	0.028	0.026–0.029	0.014	0.001	0.013	0.027	0.036	0.015
Home cap	0.020	0.017–0.023	0.009	0.000	0.011	0.018	0.042	0.009
Voters per region	0.031	0.028–0.033	0.015	0.005	0.011	0.029	0.070	0.021
Tie-break	0.002	0.000–0.003	0.000	0.000	0.002	0.000	0.001	0.000
Vote curve	0.244	0.233–0.254	0.205	0.029	0.010	0.409	0.437	0.233
Home bias	0.006	0.003–0.008	0.000	0.000	0.005	0.000	0.033	0.001
Dummy	0.003	0.002–0.004	0.000	0.000	0.003	–	–	0.000
(c) Published top 10 kept in the top 10								
Counting rule	0.105	0.093–0.117	0.018	0.002	0.085	0.036	0.045	0.019
Ballot length	0.064	0.048–0.080	0.005	0.001	0.058	0.011	0.014	0.007
Home cap	0.066	0.054–0.077	0.008	0.000	0.058	0.016	0.034	0.009
Voters per region	0.107	0.095–0.119	0.015	0.000	0.092	0.030	0.048	0.014
Tie-break	0.005	0.000–0.017	0.000	0.000	0.005	0.000	0.000	0.000
Vote curve	0.177	0.163–0.191	0.051	0.004	0.123	0.102	0.107	0.054
Home bias	0.019	0.004–0.034	0.000	0.000	0.019	0.001	0.016	0.001
Dummy	0.007	0.000–0.018	0.000	0.000	0.007	–	–	0.000
(d) Mean displacement of the published 51–100, positions below No.200 counted as No.200								
Counting rule	0.082	0.077–0.086	0.072	0.008	0.002	0.144	0.166	–
Ballot length	0.029	0.027–0.032	0.028	0.001	0.000	0.056	0.064	–
Home cap	0.001	0.000–0.001	0.001	0.000	0.000	0.001	0.005	–
Voters per region	0.013	0.011–0.014	0.005	0.007	0.001	0.010	0.034	–
Tie-break	0.000	0.000–0.000	0.000	0.000	0.000	0.000	0.000	–
Vote curve	0.375	0.366–0.383	0.295	0.076	0.004	0.590	0.612	–
Home bias	0.003	0.002–0.004	0.003	0.000	0.000	0.005	0.013	–
Dummy	0.000	0.000–0.000	0.000	0.000	0.000	–	–	–
(e) Academy regions in the top 50								
Counting rule	0.014	0.012–0.015	0.002	0.000	0.012	0.003	0.021	0.004
Ballot length	0.011	0.009–0.013	0.001	0.000	0.010	0.003	0.005	0.002
Home cap	0.040	0.037–0.044	0.007	0.017	0.016	0.015	0.089	0.027
Voters per region	0.372	0.362–0.382	0.285	0.068	0.019	0.570	0.686	0.352
Tie-break	0.002	0.001–0.004	0.000	0.000	0.002	0.000	0.000	0.000
Vote curve	0.018	0.016–0.020	0.002	0.002	0.014	0.004	0.014	0.006
Home bias	0.049	0.046–0.053	0.017	0.019	0.014	0.033	0.104	0.039
Dummy	0.004	0.000–0.008	0.000	0.000	0.004	–	–	0.000
(f) European restaurants in the top 50								
Counting rule	0.016	0.011–0.020	0.002	0.001	0.013	0.005	0.009	0.003
Ballot length	0.014	0.009–0.019	0.002	0.000	0.013	0.003	0.005	0.002
Home cap	0.019	0.015–0.023	0.002	0.003	0.014	0.004	0.029	0.005

Factor	ι_i	95% interval	Location	Spread	Shape	S_i	S_{Ti}	ι_i smoothed
Voters per region	0.128	0.119–0.138	0.106	0.006	0.017	0.211	0.247	0.108
Tie-break	0.004	0.000–0.007	0.000	0.000	0.004	0.000	0.000	0.000
Vote curve	0.017	0.013–0.020	0.000	0.004	0.012	0.001	0.005	0.005
Home bias	0.019	0.015–0.023	0.003	0.001	0.015	0.006	0.024	0.004
Dummy	0.006	0.001–0.011	0.000	0.000	0.006	–	–	0.000

References

Emanuele Borgonovo, Alessio Figalli, Elmar Plischke, and Giuseppe Savaré. Global sensitivity analysis via optimal transport. *Management Science*, 71(5):3809–3828, 2025. doi: 10.1287/mnsc.2023.01796.